BAYU AJI TRISASONGKO

B.S. in Electrical Engineering (Telkom University) · Vocational Mechatronics (SMK N 7 Semarang)

■ Semarang & Bandung, Indonesia | ✉■ bayuajitsasongko@gmail.com | ■ +62 812-2575-3129 | ■ linkedin.com/in/bayuajitrisasongko

■ CLICK HERE TO OPEN INTERACTIVE WEB PORTFOLIO & VIDEO DEMOS

https://portobayu.vercel.app — Features physical robot test videos, 3D CAD modals, and quantitative test data

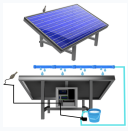
SUMMARY OF 5 ENGINEERING PROJECTS (HARDWARE, EMBEDDED & MECHATRONICS)



[01] 5-DOF Robotic Arm Manipulator (Bachelor Thesis)

Hospital service robot arm featuring 3D-printed PLA cycloidal gearboxes. Closed-loop control via STM32F411 MCU and 12-bit AS5600 magnetic encoders.

■ **Accuracy: 79.8% · Validated Payload: 250–1,000 g · IK Error: 6.65 mm** | Watch Video & Case Study >



[02] Automated Solar Panel Cooling (Capstone Design)

Automated water-spray cooling system driven by Arduino Uno with waterproof thermal probe. Actuates pump at >30°C and shuts off at <25°C with off-grid battery autonomy.

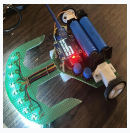
■ **Thresholds: >30°C / <25°C · Flow: 4.02 L/min · Temp Drop: >16°C** | Watch Video & Case Study >



[03] Competition Basketball Robot (National Finalist KRAI 2025)

Dual-action pneumatic dribble subsystem (5–6 bar) and tri-roller high-RPM flywheel shooter powered by 3x RS775 DC motors on holonomic 3-wheel omni base.

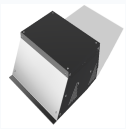
■ **Shooting Range: 6–7 Meters · Pressure: 5–6 Bar · 3x RS775 Flywheel** | Watch Video & Case Study >



[04] Autonomous Line Follower Robot (Independent Build)

High-speed line tracker with curved 8-photodiode sensor bar, 100 Hz discrete PID steering on Arduino Nano, TB6612FNG MOSFET driver, and I2C OLED telemetry.

■ **8 Photodiode Bar · Closed-Loop PID · I2C OLED Telemetry** | Watch Video & Case Study >



[05] 1 kg Bluetooth Sumo Robot 4WD (Independent Build)

20×20 cm combat robot powered by 4x high-torque PG24 planetary motors, low-angle acute front wedge scoop (<25°), 3S LiPo battery, and Bluetooth telemetry.

■ **1 kg / 20×20 cm Limit · 4WD PG24 Planetary · Bluetooth RC** | Watch Video & Case Study >

VERIFIED INDUSTRY CERTIFICATIONS & CREDENTIALS

- **FESTO & TÜV Rheinland:** Industrial Pneumatic Maintenance Technician (Cert No: 62372/21/1185)
- **National Finalist KRAI 2025:** ABU Robocon Indonesia (Puspresnas / Belmawa Kemdiktisaintek & ITB Jatinangor)
- **PT Promanufacture Indonesia (Formulatrix):** 2x Internships (Process Eng 2025 SMT/WI & Mechanical Eng 2021)
- **Telkom University:** English Proficiency Test (EPrT Score 450) & STAR Career Coach (Intrapreneurship)